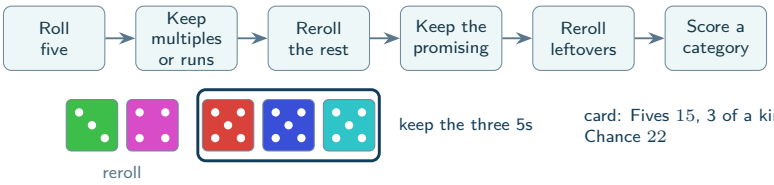
YAHTZEE ON ONE PAGE

Yahtzee: the probability of each hand

Roll, keep, reroll, keep, reroll, then score: counting turns a game of luck into a choice of odds.

1 / THE RULE AND THE PICTURE

Five dice, three rolls, and a choice of category at the end



One roll has $6^5 = 7776$ equally likely outcomes, so

$$P = \frac{\text{favourable outcomes}}{7776}$$

and rerolls act only on the free dice, so probabilities multiply along a tree and add across branches.

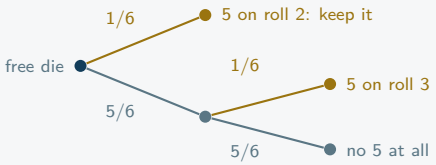
2 / ONE ROLL / COUNTING EACH HAND AMONG THE 7776

Choose the face, choose the places, then fill the other dice

hand	workings	outcomes	probability
Yahtzee	6 faces	6	0.077%
four of a kind (exactly)	$6 \times 5 \times 5$: the face, the odd die's place, its value	150	1.93%
full house	$6 \times 5 \times \binom{5}{3}$: triple face, pair face, places of the triple	300	3.86%
three of a kind (exactly)	$6 \times \binom{5}{3} \times 5 \times 4$	1200	15.43%
two pair	$\binom{6}{2} \times \frac{5!}{2!\,2!} \times 4$	1800	23.15%
one pair	$6 \times \binom{5}{2} \times 5 \times 4 \times 3$	3600	46.30%
five different	$6 \cdot 5 \cdot 4 \cdot 3 \cdot 2$	720	9.26%
small straight (a run of 4)	$3 \times 480 - 240$ for the runs counted twice	1200	15.43%
large straight	$2 \times 5!$	240	3.09%

3 / KEEPING AND REROLLING / THE DICE ON THE SCREEN

With $3, 4, 5, 5, 5$ keep the three 5s: each free die gets two chances



A free die ends as a 5 with probability $\frac{1}{6} + \frac{5}{6} \cdot \frac{1}{6} = \frac{11}{36}$, and the two free dice are independent, so keeping every 5 and rerolling the rest gives:

five 5s, a Yahtzee	$(11/36)^2 = 121/1296$	9.3%
exactly four 5s	$2 \cdot (11/36)(25/36) = 550/1296$	42.4%
exactly three 5s	$(25/36)^2 = 625/1296$	48.2%
four of a kind or better	$671/1296$	51.8%

4 / CHOOSING A CATEGORY / PROBABILITY TIMES POINTS

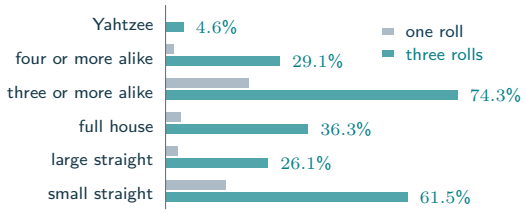
The same five dice support several plans, and expected score ranks them

aim	keep	P(made)	expected score
Yahtzee, 50 points	the three 5s	9.3%	4.7
four of a kind	the three 5s	51.8%	12.3
full house, 25	the three 5s first, then best play	30.6%	7.6
large straight, 40	$3, 4, 5$	27.8%	11.1
small straight, 30	$3, 4, 5$	80.2%	24.1
Fives	the three 5s	always	18.1
3 of a kind or Chance	the three 5s, reroll low dice	always	23.5

The card's 22 is close to the best figure here, so the choice turns on which boxes are still empty.

5 / A WHOLE TURN / BEST PLAY FOR ONE TARGET

Keeping dice lifts every hand, most of all the rare ones



Choosing the best keep at every step for one target, by searching all subsets of the five dice, gives 4.6% for a Yahtzee in a turn against 0.077% in one roll, and 26% for a large straight against 3.1%. Rare hands gain most in ratio, since a kept die is a success that need not be earned twice.

ASIDE / WHY KEEPING WORKS

A chased die succeeds within r rolls with probability $1 - (\frac{5}{6})^r$: $\frac{1}{6}$, $\frac{11}{36}$, $\frac{91}{216}$.

ASIDE / CHECK IN PYTHON

A memoised search over the 252 multisets of five dice, trying every kept subset, returns 0.046029 for Yahtzee and `(11/36)**2` gives 0.0934.

Convention. Dice and card follow the screenshot: three 5s held, two dice free. Fair dice, straights counted as runs, and best-play figures aim for one hand alone, ignoring the other categories.

Constructed by Silicon · Anthropic Claude — Architect Carbon · LeeMcCullochJames